

Jason White | Roboticist

☎ 510-359-8681 🌐 jw18dn.github.com 🔥 jwhite5497 🐙 jw18dn

Employment

Foundation Futures Industries

2026 - present

Senior Controls and Software Engineer (C++ & Python)

- o Lead controls and software development for a 22-DOF tendon driven hand, scaling it from a single motor prototype to a modular control stack
<https://www.youtube.com/watch?v=Nhr7ZnFZYkA&t=1s>
- o End to end deployment of hand hardware bring up to mounting and control on Phantom robot using RL policy whole body control with motion retargeting and mocap tracking as the system input
- o Led design decisions on tactile & position sensing, PCB components & architecture, and communication protocol & rate
- o Built full stack codebase and control of simulation and hardware in **eight weeks**
- o Developed calibration routine, system identification routine, and safety control in **three weeks**
- o Created highly accurate open loop position and torque dependent state estimator with a 10 microsecond solve time in **two weeks**

Ghost Robotics

2024 - 2026

Controls and Software Engineer (C++ & Python)

- o Built a whole-body MPC locomotion controller on Pinocchio rigid body dynamics, solving at 1kHz on onboard compute
- o Led sys ID and profile design of next-generation leg actuators and drive train, delivering **6x torque density** (3x peak torque and half the mass)
- o Cut falls & flips on rugged terrain from **35% to 5%** over 20 trials by expanding the state machine and tuning heuristics
- o Developed point-cloud clustering and 2D height-map filtering for terrain-aware locomotion, enabling traversal of grated surfaces and grated stairs
- o Integrated a deterministic finite state machine for blind locomotion, eliminating low-level crashes across months of subsequent field testing
- o Designed filtering and tracking pipelines for quadruped mounted 6 DOF manipulator, removing drift and reducing end-effector tracking error

Optimal Robotics Lab

2018 - 2023

Graduate Research Assistant (Matlab & Python)

- o Obstacle avoidance algorithm: for quadrotor & bipedal avoidance of fast-moving obstacles
[hardware, ~1kHz] <https://www.youtube.com/watch?v=cKKLuRdrqcQ>
- o Bipedal balancing controller: for dynamic terrain tested on Cassie
[hardware, >2kHz] <https://www.youtube.com/watch?v=q10Wg61H4UQ>
- o Real-time reactive planning and control for monopods (with flexible contact sequencing)
[simulated, ~500Hz] <https://www.youtube.com/watch?v=EquyU9cm-SM>
- o Quadrupedal stepping controller: using optimal control techniques (preliminary results)
[simulated, ~500Hz] www.linkedin.com/feed/update/urn:li:activity:7122585426162249730/

Technical Skills

Languages: C++, Python, MATLAB

Simulations: MuJoCo, Simulink, Bullet

Control Methods: Operational Space Control, Whole-Body Control, Forward/Inverse Dynamics, Joint Space Control, PD/PID Control, Force Control

Planning: Model Predictive Control, Direct Collocation, Sampling-based methods (RRT*, A*)

Optimization: Linear/Quadratic/Nonlinear Programming, Newton-Raphson

Awards, Fellowships, and Honors

2022: NASA Big Idea Challenge Finalist (ET-QUAD)

NASA

2022: Best Locomotion Paper, IEEE International Conference on Robotics and Automation

IEEE

2021: Research Fellowship Recipient, DEVCOM Army Research Laboratory

National

Peer Reviewed Publications

Tianze Wang, **Jason White**, and Christian Hubicki, "Real-time Dynamic Bipedal Avoidance," in 2023 *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)* IEEE.

Jason White, David Jay, Tianze Wang, and Christian Hubicki, "Avoiding Dynamic Obstacles with Real-time Motion Planning using Quadratic Programming for Varied Locomotion Modes," in 2022 *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)* (pp. 13626-13633) IEEE.

Adwait Mane, Dylan Swart, **Jason White**, and Christian Hubicki. "Trajectory Optimization Formulation with Smooth Analytical Derivatives for Track-leg and Wheel-leg Ground Robots." in 2022 *International Conference on Robotics and Automation (ICRA)* (pp. 5762-5768). IEEE.

Jason White, Max Austin, Jonathan Clark, Christian Hubicki, and Jason Pusey. "Online Gait Optimization for Running in Resistive Media," in 2022 *7th International Conference on Robotics and Automation Engineering (ICRAE)* (pp. 282-286). IEEE.

Max Austin, John Nicholson, **Jason White**, Sean Gart, Ashley Chase, Jason Pusey, Christian Hubicki, Jonathan Clark, "Optimizing Dynamic Legged Locomotion in Mixed, Resistive Media," in 2022 *IEEE/ASME International Conference on Advanced Intelligent Mechatronics (AIM)* (pp. 1482-1488). IEEE.

Jason White, Dylan Swart, and Christian Hubicki, "Force-based Control of Bipedal Balancing on Dynamic Terrain with the "Tallahassee Cassie" Robotic Platform," in 2020 *IEEE International Conference on Robotics and Automation (ICRA)* (pp. 6618-6624). IEEE.

Manuscripts & Posters

Jason White and Christian Hubicki, "Through-contact Monopod Motion Generation via MPC: Convex Formulation and Locomotion Analysis." 2024.

Jason White and Christian Hubicki, "Real-time Adversarial Obstacle Avoidance." 2023.

Jason White and Christian Hubicki "Toward Reduced-Order Modeling Approaches for Rapidly Reactive Locomotion." 2021 Dynamic Walking

Education

Florida State University *Advisor: Christian Hubicki Ph.D.*

Ph.D. Mechanical Engineering - Legged Robot Control

M.S. Mechanical Engineering - Robotics, Dynamics, & Controls

Virginia Polytechnic Institute and State University

B.S. Civil Engineering - Structural Focus